MILITARY SPECIFICATION

REPEATER, MULTI-SPEED, 3-CHANNEL, NAVIGATION SYSTEM

1. SCOPE

1.1 This specification covers the multi-speed repeater for use in the navigation subsystem. This repeater (similar to a synchro-amplifier) reads out inductosyn and resolver orientation or inertial navigators, and retransmits these angular data at various speeds and data forms as required for system and general shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-C-2194 - Cables, Power, Electrical, Reduced Diameter Type, Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Rarts, Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OP-1303 - U. S. Navy Synchros: Description and Operation.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be inspected as specified in 4.2 (see 6.2).

3.2 Description. - Each channel of the multi-speed repeater (one channel is shown schematically on figure 1), shall use a high accuracy 10 kilocycle (kc.) resolver mounted on a shaft or integral with a torque motor or other suitable type motor. The resolver is excited (at about 30 volts, 10 kc.) and transmits sinecosine voltages to an external inductosyn mounted on an inertial system. If the inductosyn (a very precise

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special multi-speed resolver) is not at the same electrical setting as the resolver, an error or difference signal is developed across the inductosyn. This error signal is then suitably processed and used to control the rotation of the torque motor. The resolver is then repositioned so that it is at the same setting as the inductosyn and hence no error signal is developed. A similar servo channel exists at one speed to receive rough positioning signals from a resolver transmitter (one speed) on the same axis as the inductosyn to control the torque motor in the repeater. Switch-over circuits shall be included to allow use for the one-speed or multi-speed signals to drive the torque motor. A multiplicity of transmitters at various speeds are geared to the one speed and multi-speed shafts.

3.3 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant material
Toxic materials
Wood
Aluminum
Magnesium
Iron or steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Locking devices
Washers
Ball bearings (precision grade required)
Parts - electrical - general
Requirements for semiconductor devices
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers and inductors
Relays
Synchros
Electrical tapes
Dial illumination lamps
Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuse clips
Metallic rectifiers
Printed wiring and circuits
Enclosures - general

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Enclosures - accessibility
Enclosures - degree of (dripproof) (see 3.4.5)
Enclosure - mounting
Stiffening grooves
Minimum sheet metal thicknesses (watertight enclosures)
Through bolting
Cable entrance
Ventilation
Size (for submarines)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity
Accelerated life
Shock, vibration and inclination
Primary power supply circuits
Power supply tolerances
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (test cable required)
Terminal boards, connectors and terminals
Synchro connections and markings
Wiring
Dial illumination
Electrical insulation
Dielectric strength and insulation resistance clearances
Airborne noise (see 3.4.17)
Structureborne noise (see 3.4.17)
Drawings - general
Drawings - preliminary
Drawings - working
Schematic diagrams
Wiring diagram
Assembly drawings
Drawings - manufacturing
Bills of materials
Interchangeability and standardization
Manuals
Repairs parts
Designation and marking (serial numbers are required)
Workmanship and general examination

3.4 Detailed features. -

3.4.1 Input information is from an external one-speed, 400-cycle resolver and from an external 180-speed, 10 kc. inductosyn (for each of the three channels for roll, pitch and heading).

3.4.1.1 The rotor of the external one-speed, 400-cycle resolver shall be excited by a 26-volt, 400-cycle reference voltage from the repeater, and shall transmit four-wire, two-signal information to the repeater, with a gradient of 0.4 volt per degree (±10 percent), maximum amplitude of 26 volts and null of about 26 millivolts. The 26-volt 400-cycle reference voltage source in the repeater shall be amply sized to provide the fully loaded excitation power requirements of the three external one-speed resolvers supplying all three channels with signal inputs. For the roll and pitch channels the external resolver is the Kearfott type RS931-3; for the own ship's heading channel the external resolver is the Kearfott type CR40931030.

3.4.1.2 The 180-speed external inductosyn circuit shall be provided with four-wire, two-signal resolver information with a maximum amplitude of five volts at 10 kc. from the repeater. The error signal returned to the repeater will be as indicated in 3.4.16 and 3.4.18.

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3.4.2 Output is at 1-, 2- and 36-speed, 400-cycle synchro for each channel. Size 23Cx4 synchros are required (two for each speed).

3.4.2.1 The three channels for roll, pitch, and heading shall be interchangeable.

3.4.2.2 Dial indicators are required at 1-speed and 180-speed for each channel.

3.4.3 Accuracy.- The overall error of multi-speed repeaters shall not exceed the values shown in table I under all test conditions specified in 4.4.

Table I - Accuracy

Condition	Error in minutes referred to 1-speed shaft			
	Input speed	Output Speed		
	1 and 180	1	36	2
Static	1 and 180	12.0	1.0	6.0
Dynamic: Maximum velocity 1/ 18 degrees per second or simple harmonic motion of 35-degree amplitude and 12 second period.	1 and 180	12.0	1.0	6.0
Maximum acceleration 1/ 10 degrees per *second squared*.	1 and 180	12.0	1.0	6.0

1/ Velocity and acceleration refer to motion of one speed shaft.

3.4.4 The repeater shall be capable of being tested and maintained onboard ship by Naval personnel. Construction shall permit easy disassembly, repair or substitution by sub-units. Circuits shall be properly segregated to minimize effects of malfunctions and adequate test points provided for quick troubleshooting.

3.4.5 Enclosure.- The three channels of the multi-speed repeater and a 10 kc. power supply shall be grouped together in one dripproof aluminum enclosure, suitable for four-point deck mounting and two-point bulkhead support. Front access for maintenance of all internal components is required. All gears shall be protected from dust. A removable plate is required on the back of the enclosure for installation of stuffing tubes for cable entrance.

3.4.5.1 The plate shall provide for two cables (one cable being a special cable about 1-inch in diameter and the other being a type 2SA cable) to connect to the external signal source and for necessary standard Navy type cables of MIL-C-2194 to connect to the ships power, to the external alarm circuit and to the outputs of the repeater.

3.4.6 Size.- The repeater group with power supply shall be as compact as is consistent with number of components, accuracy required and ease of maintenance. Overall size of the enclosure shall be within the size limitations for submarine installations in MIL-I-983.

3.4.7 Shock mounts.- External shock mounts may be used, if necessary, to insure meeting the shock and vibration requirements of submarine installations.

3.4.8 Power input.- The repeater shall operate on 115-volt, 400-cycle, single-phase or balance 3-phase ship's service power. A 10-kc., 30-volt supply with frequency stability of 1/2 percent shall be furnished within the repeater assembly and used as a common supply for all three channels. Circuits within the repeater shall provide 26-volt, 400-cycle excitation to the external 1-speed transmitting resolver supplying each channel.

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3.4.9 Provision shall be made for conveniently zeroing the inductosyn electrical circuit without affecting other units. Input and output resolvers and synchros shall be correctly zeroed and sequenced with reference to ship's wiring connection terminals. Synchros zeroing shall be in accordance with Publication OP-1303. Resolver zeroing and phase sequence shall be in accordance with U. S. Naval Applied Science Laboratory Technical Report "Inductosyn Transmission Systems." Use of resolvers other than shown therein shall be approved by the Bureau of Ships. (Copies of aforementioned report are available from Bureau of Ships, Code 665.)

3.4.10 Electron tubes. - Electron tubes shall not be used. Solid state devices shall be used throughout. Standard Navy approved components shall be used wherever possible.

3.4.11 Indicator lights. - Separate indicator lights having white lenses shall be provided to give a single indication of failure in the following:

(a) Servo system follow-up action of each channel (total of three lights required).
(b) Input power (one light required).
(c) Output 400-cycle synchro excitation (one light required).

All lights shall be connected so as to be lit for normal operation of the repeater and to go off individually for failures.

3.4.12 Alarm relay and switch. - An alarm relay shall be provided for the three channels taken collectively to indicate failure of the servo system to follow the input signals, loss of input excitation or loss of output synchro excitation. The relay contacts shall be designed to carry 5 amperes at 0.5 power factor, 115-volts, 60 or 400-cycles a.c., and shall be connected to an external alarm circuit. The relay shall operate when the servo error exceeds plus or minus 30.0 minutes, or upon loss of input power or output synchro excitation. An alarm relay switch shall be provided, and shall be connected so that the alarm relay will operate the external alarm circuit as shown in table II and on figure 2.

Table II - Alarm relay and switch connection.

Inputs and output synchro excitation	Alarm switch	External alarm circuit
Off	Off	De-energized
On	Off	Energized
On	On	De-energized
Off	On	Energized

3.4.13 The equipment shall not be adversely affected by magnetic fields up to 5 gausses.

3.4.14 Synchronization. - The equipment, within capability of input signals shall be self-synchronous from all displacement out of synchronism with the input signals.

3.4.15 Oscillations. - There shall be no spurious oscillation in the overall error as measured under any conditions covered in this specification having a total excursion greater than one quarter the allowable error (see table I) appearing in three or more consecutive cycles.

3.4.16 Phase shift and gain. - The repeater shall be provided with suitable phase adjustment to compensate for a phase shift of $-110^{\circ} \pm 20^{\circ}$ between the primary and secondary voltages of the external inductosyn circuit. It shall also be provided with error gain adjustments that will not affect phase adjustments.

3.4.17 Noise. - Noise generated by the repeater shall not exceed the grade C airborne and type 3 structureborne noise limitation in accordance with MIL-STD-740.

3.4.18 The repeater amplifier shall be capable of operating from the output of an inductosyn preamplifier having the following characteristics:

(a) 750 microvolts per second of arc ±10 percent gradient.
(b) Noise level - less than 6 millivolt at 10 kc. with higher values at 180, 400, 800, and 2400 cps.
(c) Output impedance 100 $\angle +0$ to 18° ohms maximum.
(d) External line load 5000 $\angle +0$ to 18° ohms minimum.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The supplier must obtain Bureau of Ships approval of the test equipment which will be used.

4.2 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests specified in table III. Inspection shall be performed in general in the order listed. A preproduction test procedure, in detail, shall be submitted for the approval of the Bureau of Ships. Two copies of the preproduction test results shall be submitted for the approval of the Bureau of Ships.

Table III - Preproduction inspection.

Examination and tests	Requirement paragraph	Test reference
General examination	3.3	MIL-I-983
Radio frequency noise reduction 1/	3.3	MIL-I-983
Airborne and structureborne noise	3.3 and 3.4.17	4.4.2
Synchronization	3.4.14	4.4.3
Accelerated life (Includes tests for supply line voltage and frequency variation, vibration, accuracy, temperature and humidity, dielectric strength, insulation resistance, and shock)	3.3	4.4.7
Enclosure	3.3 and 3.4.5	MIL-I-983
Effect of magnetic fields	3.4.13	4.4.4
Oscillations	3.4.15	4.4.5
Alarm operation	3.4.12	4.4.6
Synchro connections	3.3	4.4.8
Resolver connections	3.4.9	4.4.9
Phase shift	3.4.16	4.4.10

1/ The multispeed repeater shall be operating the same as specified in 4.4.2.

4.3 Quality conformance inspection. - All equipment on the contract or order shall be subjected to the group A and B examination and tests shown in table IV. Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be subjected to the group C tests shown in table IV.

Table IV - Quality conformance inspection.

Examination and tests	Requirement paragraph	Test reference
Group A		
General examination	3.3	MIL-I-983
Dielectric strength 1/	3.3	MIL-I-983
Insulation resistance 1/	3.3	MIL-I-983
Accuracy	3.4.3	4.4.1
Synchro connections	3.3	4.4.8
Resolver connections	3.4.9	4.4.9
Group B		
Oscillations	3.4.15	4.4.5
Synchronization	3.4.14	4.4.3
Alarm operation	3.4.12	4.4.6
Phase shift	3.4.16	4.4.10

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Table IV - Quality conformance inspection (Contd.)

Examination and tests	Requirement paragraph	Test reference
Group C		
Radio frequency noise reduction 2/	3.3	MIL-I-983
Airborne and structureborne noise	3.3 and 3.4.17	4.4.2
Accelerated life (Includes tests for supply line voltage and frequency variations, vibrations, temperature and humidity, dielectric strength, insulation resistance, and shock)	3.3	4.4.7
Enclosure	3.3. and 3.4.5	MIL-I-983
Effect of magnetic fields	3.4.13	4.4.4

1/ For all but rotating components of equipment

2/ The multi-speed repeater shall be operating the same as specified in 4.4.2.

4.4 Test procedures. -

4.4.1 Accuracy. -

4.4.1.1 Overall accuracy. - The overall accuracy of multi-speed repeaters shall be determined by operating the equipment with a 1- and 180-speed input from a dummy director such as is shown on figure 3 or from a suitable equivalent resolver input device. An inductosyn and resolver when used shall be mounted on the same shaft to furnish dummy director inputs to the multi-speed repeaters for purposes of determining overall accuracy. A suitable schematic test set-up is shown on figure 3.

4.4.1.1.1 Static accuracy. - The overall static accuracy for each of the three channels of the equipment shall be determined by turning the dummy director slowly in one direction in steps of 10.25 degrees of the 1-speed resolver and observing the maximum error at each step through one complete revolution of the 1-speed resolver. The same procedure shall be repeated in the opposite direction of rotation. The maximum overall error as indicated by the voltage across the rotor leads of the CT4 in the dummy director or equivalent input resolver input device shall not exceed the value specified in 3.4.3 for 1, 2 and 36-speed.

4.4.1.1.2 Dynamic accuracy. - The overall dynamic accuracy for each of the three channels shall be determined by operating the dummy director under the following conditions. All motions refer to the motion of the 1-speed input:

(a) Constant velocity 18 degrees per second in a clockwise direction; in a counterclockwise direction.
(b) Simple harmonic motion 35-degree amplitude 12-second period.

The dynamic error as indicated by the voltage across the rotor leads of the 36-speed CT4 in the dummy director shall not exceed the values specified in 3.4.3.

4.4.2 Noise. - Tests shall be made to ensure that generated noise does not exceed the limits of 3.4.17. Tests shall be conducted as specified in MIL-STD-740. For both the airborne and structureborne noise tests one channel of the repeater shall be operating with a constantly varying input signal of simple harmonic motion of 35-degree amplitude and 12-second period while remaining two channels shall be operating with static inputs at 1 and 180-speed. All output synchros shall be energized and one output synchro shall be loaded with an actual or simulated synchro load during the tests.

4.4.3 Synchronization. - Tests shall be made on each channel to ensure that the multi-speed repeater will be self-synchronous as specified in 3.4.14.

4.4.4 Effect of magnetic fields. - Tests shall be made on all three channels to ensure that a magnetic field will not affect the accuracy of the multi-speed repeater as specified in 3.4.13.

4.4.5 Oscillations. - Tests shall be made on all three channels to ensure that oscillations in the overall error do not exceed the requirements of 3.4.15.

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4.4.6 Alarm operation.- Tests shall be conducted to ensure that the alarm circuit of each channel operates as specified in 3.4.12.

4.4.7 Accelerated life.- Before this test is undertaken the repeater shall have passed the full accuracy test of 4.4.1 The accelerated life test shall be conducted as specified in MIL-I-983 with a constantly varying input signal of simple harmonic motion of 35-degree amplitude and 12-second period to one channel and static or dynamic inputs at 1 and 180-speed to the other two channels except where indicated otherwise herein. For the temperature test all output synchros of the repeater shall be energized. For this test the equipment shall be mounted between heat dissipating surfaces which will simulate a stack-mounted installation. Also the heat dissipated from the sides and back of the equipment shall be restricted by means of low heat conducting bulkheads placed adjacent to the sides and back of the equipment. The high temperature portion of the test shall be conducted at 149°F. in lieu of 131°F. Any time during the accelerated life test the equipment shall be capable of meeting the accuracy limits of 3.4.3 as measured on the 36-speed synchro for the above input conditions. Accuracy readings shall be made on all three channels during the accelerated life test. For the supply line voltage and frequency variation test each channel shall be tested for accuracy by using the static test of 4.4.1.1.1 for one direction of rotation only at approximately 10 widely distributed steps over one revolution of the 1-speed input resolver and by using the simple harmonic motion test of 4.4.1.1.2; all measurements being made on the 36-speed synchro. For the shock and vital equipment vibration tests each channel shall be energized as indicated in the second sentence of this paragraph. During the vibration test and after each shock of the shock test the accuracy of each channel measured on the 36-speed synchro shall be recorded for the inputs just prescribed. After the conclusion of the vibration and shock tests collectively considered the full accuracy test of 4.4.1 shall be performed on each channel. To pass the vibration and shock tests each channel of the multi-speed repeater shall meet the accuracy requirement of 3.4.3 whenever tested for accuracy during or after the tests as specified above.

4.4.8 Synchro connections and markings.- Tests shall be conducted to ensure that synchro connections conform to MIL-I-983.

4.4.9 Resolver connections.- Tests shall be conducted to ensure that resolver connections conform to U.S. Naval Applied Science Laboratory Technical Report "Inductosyn Transmission Systems".

4.4.10 Phase shift.- Tests shall be conducted to ensure that the equipment will meet the phase shift requirements of 3.4.16.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Multi-speed repeaters.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the suppliers commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged by level A; packed by level C and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

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(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, repair parts and publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of manuals required (see 3.3).
(c) Inventory control point for repair parts (see 3.3).
(d) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 6605-N100SH)

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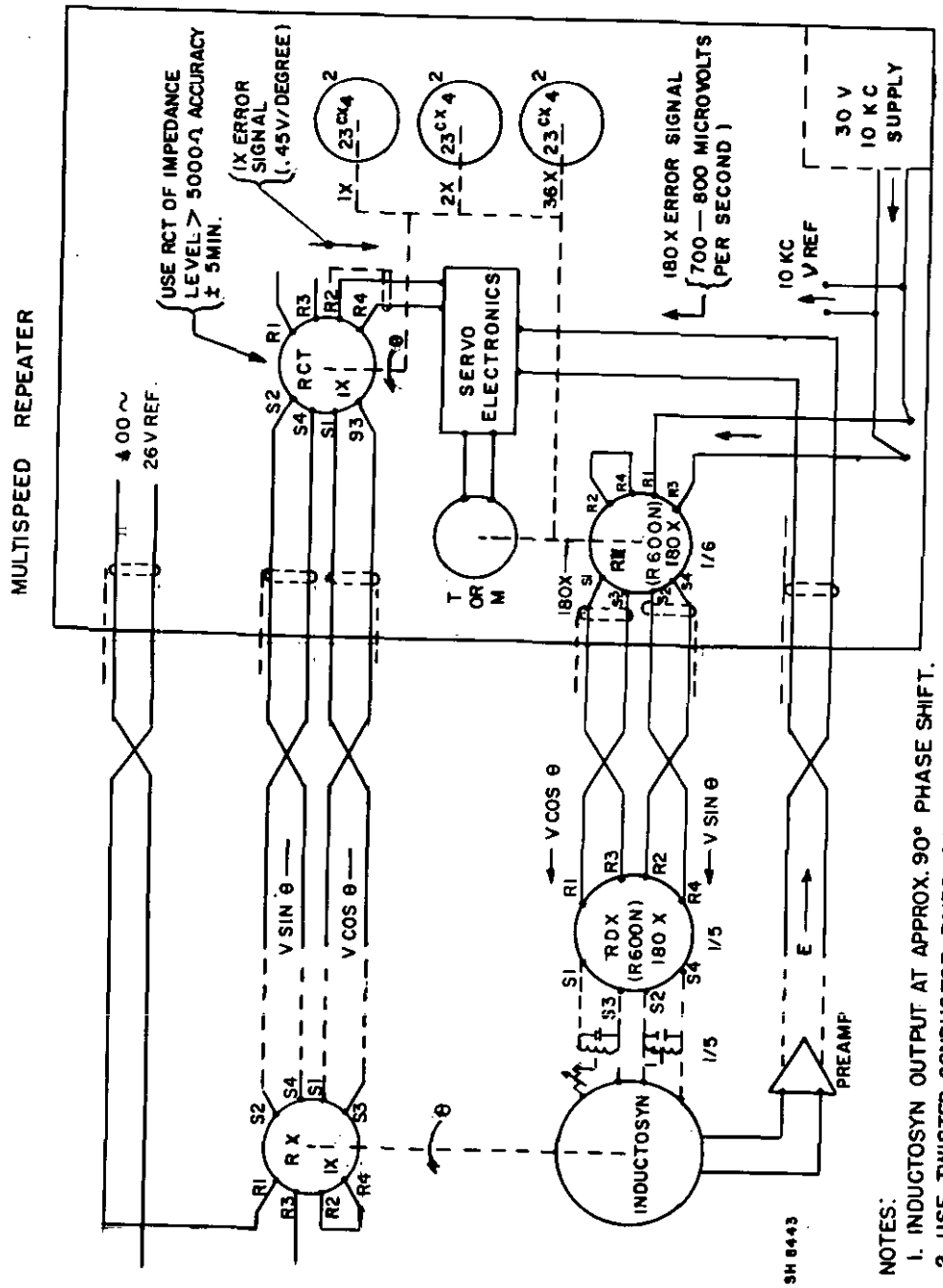


Figure 1 - One channel of multi-speed repeater.

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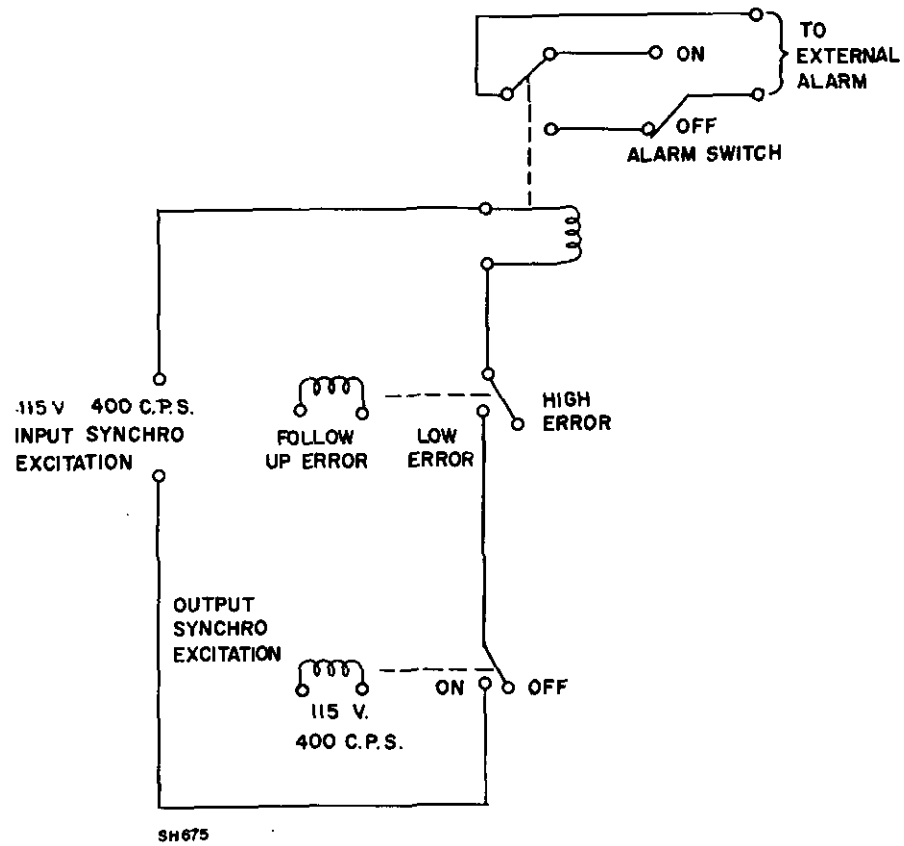
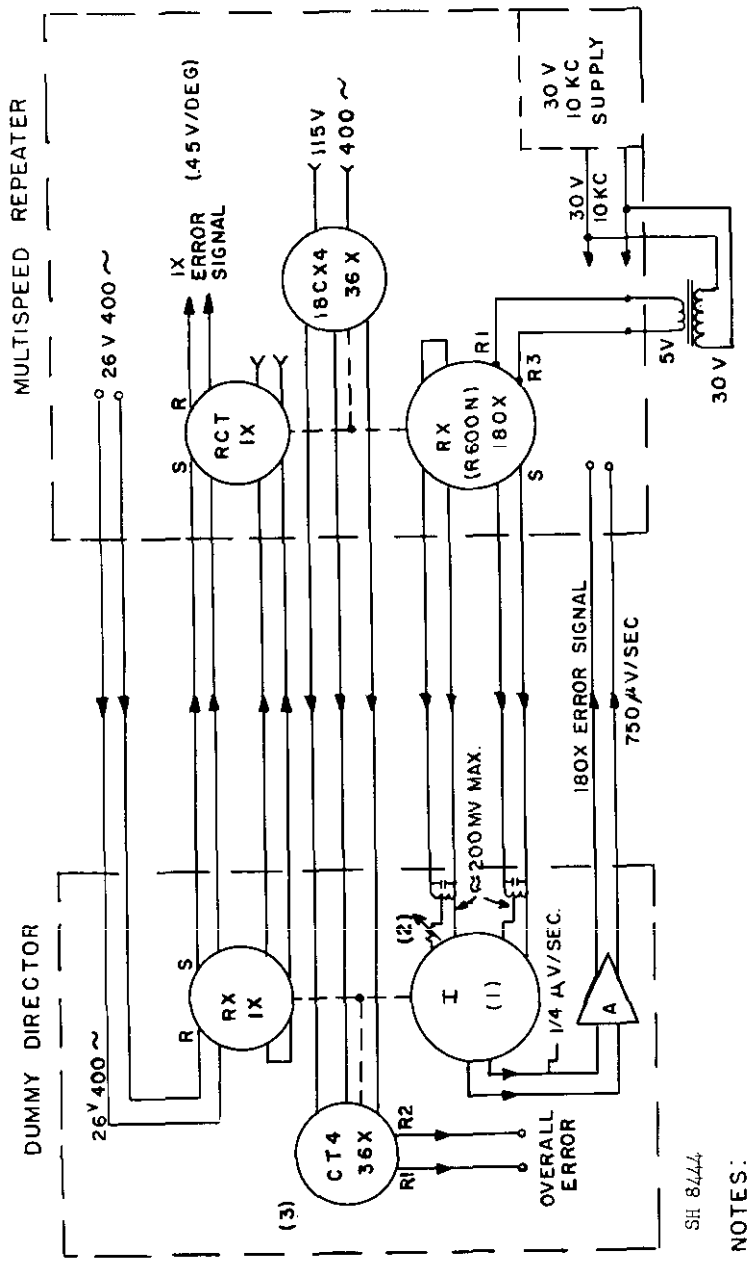


Figure 2 - Alarm circuit.

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NOTES:

1. POSSIBLY 3 INCH PACKAGED INDUCTOSYN ASSEMBLED WITH RESOLVER RX.
2. ADJUST RESISTOR TO BALANCE RESISTANCES OF INDUCTOSYN.
3. USE A 3-MINUTE SYNCHRO.

Figure 3 - Schematic test.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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